

# SECTION 7: Prompt Engineering Mastery

Welcome to the art and science of communicating with AI. Prompt engineering is the skill that separates mediocre results from excellent ones. It does not require coding knowledge but relies on clear communication, logic, and an understanding of how AI models "think."

## Lecture 7.1: Introduction to Prompt Engineering

### What is Prompt Engineering?

A **Prompt** is the input (text, image, etc.) you give to an AI model to instruct or guide it. **Prompt Engineering** is the practice of crafting these inputs to get the most useful, accurate, and relevant outputs.

#### Why It Matters

AI models are powerful but literal. They lack common sense and context unless you provide it. A small change in phrasing can dramatically alter the result because models generate text by predicting the next likely token based on the patterns in your prompt.

### Figure 1: The Impact of Specificity

#### BEFORE (VAGUE)

"Write about climate change."

*Result: Generic, unfocused overview.*

#### AFTER (SPECIFIC)

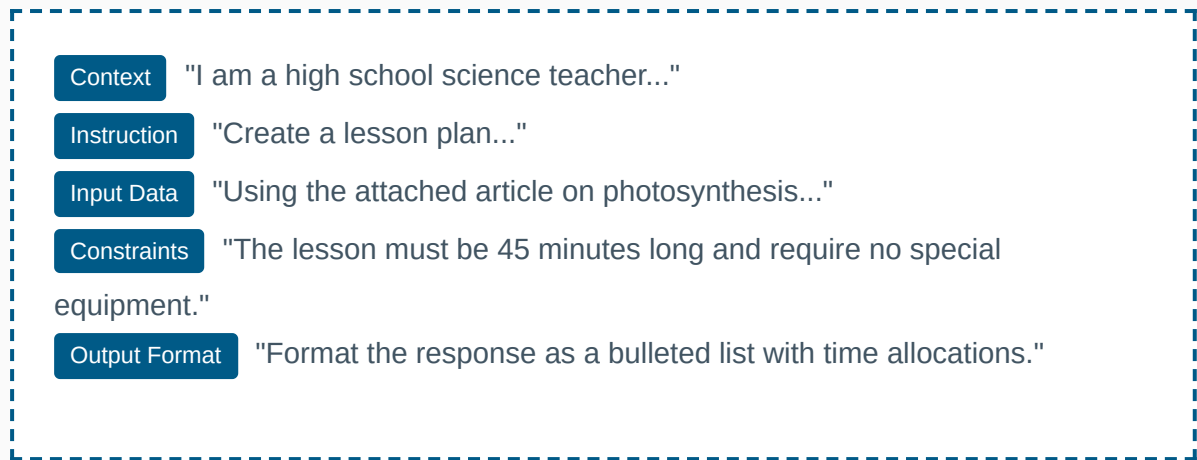
"Write a 300-word explanation of how climate change affects ocean ecosystems, suitable for high school students. Focus on coral reefs."

*Result: Tailored, relevant, and useful content.*

### Elements of a Good Prompt

While not every prompt needs all elements, effective ones typically contain:

**Figure 2: Anatomy of a Good Prompt**

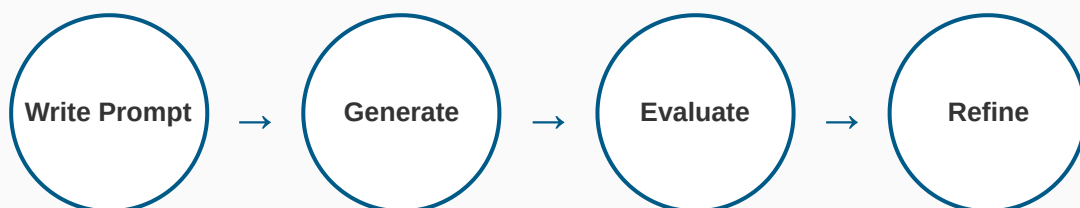


## The Science Behind It

- **Context-Awareness:** Models attend to all parts of the prompt. If you say "Explain like I'm 5," the attention mechanism focuses on simple vocabulary patterns.
- **Stylistic Associations:** Phrasing like "As an expert in..." triggers more authoritative and technical patterns found in the training data.

## The Iteration Cycle

Prompt engineering is rarely "one and done." It is an iterative process.



## Common Misconceptions

| Myth                           | Reality                                                                                                 |
|--------------------------------|---------------------------------------------------------------------------------------------------------|
| "There is one perfect prompt." | Many approaches can work; it's about finding what works best for your specific goal.                    |
| "Politeness matters."          | AI doesn't have feelings, but polite language often correlates with helpful responses in training data. |

"Longer is always better."

Conciseness and clarity often beat length. Unnecessary fluff can confuse the model.

## LECTURE 7.1 KEY TAKEAWAY

Prompt engineering is "programming in natural language." It is a fundamental digital skill that multiplies productivity by bridging the gap between human intent and AI execution.

## Lecture 7.2: Core Prompting Techniques

### 1. Zero-Shot Prompting

Asking the AI to perform a task without any examples. Best for common, straightforward tasks where the model has plenty of training data.

*Example: "Translate this sentence to Spanish: [Sentence]"*

### 2. Few-Shot Prompting

Providing 2-5 examples of inputs and desired outputs to establish a pattern. Crucial for ensuring specific formats or handling nuanced tasks.

#### Few-Shot Example:

Review: "This movie was great!" -> Sentiment: Positive

Review: "I fell asleep." -> Sentiment: Negative

Review: "It was okay, not amazing." -> Sentiment: Neutral

Review: "Best film of the year!" -> Sentiment: **[AI completes pattern]**

### 3. Chain-of-Thought Prompting

Instructing the model to show its work before giving an answer. This significantly improves performance on math, logic, and reasoning tasks.

**Magic Phrase:** "Let's think step by step."

### 4. Role Prompting

Assigning a persona to the AI to control tone, vocabulary, and perspective.

- "Act as a Marine Biologist" -> Technical, scientific, detailed.
- "Act as a Kindergarten Teacher" -> Simple, encouraging, uses analogies.

## 5. Instruction Clarity & Constraints

Be specific about what you want and what you don't want (Negative Constraints). Constraints act as guardrails.

- **Length:** "In exactly 3 sentences..."
- **Format:** "Use a markdown table..."
- **Scope:** "Focus only on the economic impacts..."

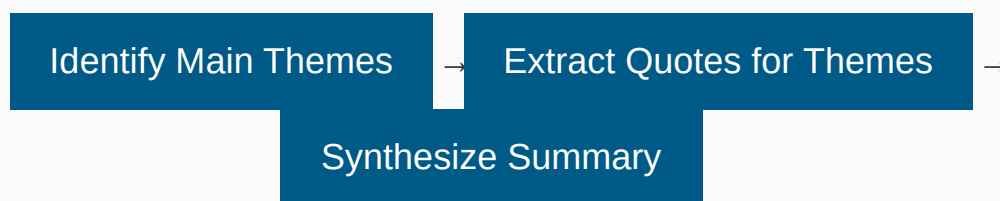
## 6. Context Provision

Giving background info (who you are, what the goal is) transforms generic advice into actionable consulting.

## 7. Decomposition

Breaking complex tasks into a sequence of smaller, simpler prompts.

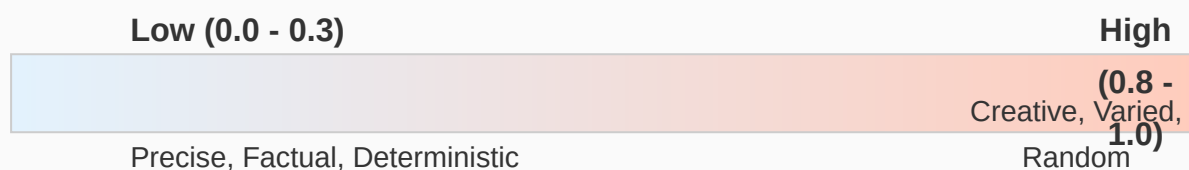
**Figure 3: Task Decomposition**



## 8. Temperature and Parameters

Controlling the randomness of the output.

**Figure 4: Temperature Scale**



## Practical Patterns

| Pattern     | Example                                                             |
|-------------|---------------------------------------------------------------------|
| Explanation | "Explain [Concept] to a [Audience] using an analogy."               |
| Analysis    | "Analyze [Topic] considering 1. Cost, 2. Time, 3. Quality."         |
| Review      | "Identify 3 strengths, 3 weaknesses, and provide a recommendation." |

## LECTURE 7.2 KEY TAKEAWAY

Master the core toolkit: Zero-shot for simple tasks, Few-shot for patterns, Chain-of-thought for logic, and Roles for tone. Combine these with clear constraints and context for powerful results.

## Lecture 7.3: Advanced Prompting Strategies

### 1. Self-Criticism and Reflection

A multi-step process where the model critiques its own work to improve quality.

**Figure 5: The Reflection Loop**



### 2. Chain-of-Verification (CoVe)

Reducing hallucinations by forcing the model to fact-check itself.

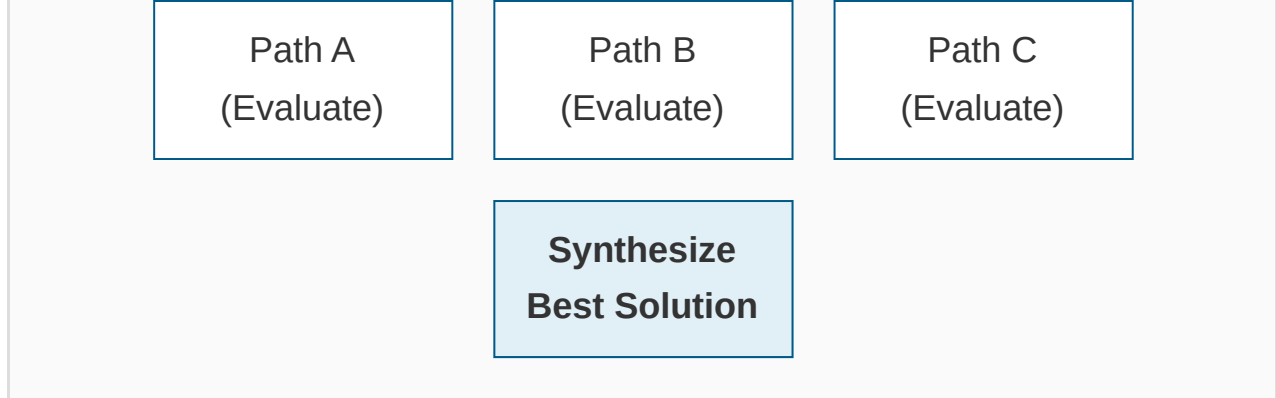
1. Generate baseline response.
2. Generate verification questions identifying potential errors.
3. Answer verification questions.
4. Produce final revised response.

### 3. Tree-of-Thoughts

Used for complex problem solving where the model explores multiple possible solutions simultaneously.

**Figure 6: Tree of Thoughts Logic**

**Problem**



## 4. Meta-Prompting

Using the AI to help you write better prompts. *"I want to do X. What is the best prompt to achieve this?"*

## 5. Retrieval-Augmented Generation (RAG) style Prompting

Simulating RAG by pasting source text into the prompt and instructing the model: "Answer ONLY using the provided text." This grounds the response and reduces hallucination.

## 6. Contrastive Prompting

Providing examples of what to do AND what NOT to do.

### GOOD EXAMPLE

"Dear Mr. Smith, I am writing to inquire..." (Professional)

### BAD EXAMPLE

"Hey, what's up with the project?"  
(Too Casual)

## Other Advanced Strategies

- **Directional Stimulus:** Using keywords (e.g., "Keywords: Witty, Optimistic") to steer the output.
- **Staged Refinement:** Improving an output in passes (Outline → Draft → Polish).
- **Perspective-Taking:** "Analyze this from the perspective of an Economist, a Sociologist, and an Environmentalist."

### ⚠ Anti-Patterns to Avoid

- **Kitchen Sink:** Putting too many complex instructions in one prompt.
- **Assuming Knowledge:** Thinking the AI knows context you haven't shared.

- **Ignoring Failures:** Accepting bad output without giving feedback.

## LECTURE 7.3 KEY TAKEAWAY

Advanced strategies move beyond simple instructions to structured reasoning workflows. Techniques like CoVe and Tree-of-Thoughts allow AI to handle complex, high-stakes tasks with greater reliability.

## Lecture 7.4: Prompt Engineering for Different Modalities

### Image Generation Prompting

Unlike text, image prompting is often "painting with words." It requires vivid visual descriptions.

**Figure 7: Image Prompt Anatomy**

|             |                                                  |
|-------------|--------------------------------------------------|
| Subject     | "A futuristic city"                              |
| Details     | "neon lights, flying cars, rain-slicked streets" |
| Style       | "Cyberpunk, Digital Art"                         |
| Lighting    | "Cinematic lighting, blue and purple hues"       |
| Composition | "Wide angle, bird's eye view"                    |
| Quality     | "8k, Unreal Engine 5 render, highly detailed"    |

### Unique Considerations:

- **Negative Prompts:** Listing what you DON'T want (e.g., "blur, text, low quality, extra fingers").
- **Weights:** Emphasizing terms (e.g., "(fog:1.5)" makes it very foggy).
- **Seeds:** Locking the random seed to reproduce the same style across variations.

## Audio & Video Prompting

| Modality | Key Elements                                                                                    | Anti-Patterns                                                                            |
|----------|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Audio    | Genre, Instrumentation, Tempo, Mood, Context (e.g., "Upbeat jazz for a coffee shop background") | Being too abstract ("Sound like the color blue") or overly specific ("Exactly 124 BPM"). |
| Video    | Subject, Action, Motion (pan, zoom), Camera Angle, Duration                                     | Complex narratives ("A man is born, grows up, and dies")—keep actions short and simple.  |

## Multimodal & Cross-Modal Prompting

Combining inputs (e.g., Image + Text) or converting modalities (Image-to-Text).

- **Image-to-Image:** "Make this photo look like a sketch."
- **Text-to-Video:** "A drone shot of a cliffside" (requires handling temporal coherence).

## Achieving Consistency

To keep characters or styles consistent across multiple generations, use **Seed Locking**, **Reference Images**, and extremely detailed, identical descriptions for the subject in every prompt.

### Building Expertise

Every model (Midjourney vs. DALL-E vs. Runway) has its own "dialect." Midjourney prefers artistic keywords; DALL-E prefers natural sentences. Mastery comes from experimenting with each platform.

### LECTURE 7.4 KEY TAKEAWAY

Prompting principles adapt across modalities. While text relies on logic and context, visual and audio prompting rely on descriptive density and style references. Consistency is the hardest challenge but can be managed with advanced controls like seeds and reference inputs.